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# CBLOF Implementation using Python

import numpy as np
import matplotlib.pyplot as plt
from sklearn.cluster import KMeans
from sklearn.datasets import make_blobs

# Generate sample data
X, _ = make_blobs(n_samples=200, centers=3, cluster_std=1.0,
random_state=42)

kmeans = KMeans(n_clusters=3, random_state=42)
labels = kmeans.fit_predict(X)

unique, counts = np.unique(labels, return_counts=True)
cluster_sizes = dict(zip(unique, counts))

threshold = np.mean(list(cluster_sizes.values()))

large_clusters = [k for k, v in cluster_sizes.items() if v >=
threshold]
small_clusters = [k for k, v in cluster_sizes.items() if v <
threshold]

cblof_scores = []

for i, point in enumerate(X):
    cluster = labels[i]

    if cluster in large_clusters:
        score = 0
    else:
        distances = []
        for lc in large_clusters:
            center = kmeans.cluster_centers_[lc]
            distances.append(np.linalg.norm(point - center))

        score = cluster_sizes[cluster] * min(distances)

    cblof_scores.append(score)

cblof_scores = np.array(cblof_scores)

threshold_score = np.percentile(cblof_scores, 90)
outliers = X[cblof_scores > threshold_score]

plt.scatter(X[:, 0], X[:, 1], c='blue', label='Normal Data')
plt.scatter(outliers[:, 0], outliers[:, 1], c='red', label='Outliers')
plt.title("CBLOF Outlier Detection")
plt.legend()
plt.show()

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